Non-Stabilizing Power of the Quantum Fourier Transform via Matrix Product Operator Representation

Working Notes

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Abstract

We develop an analytical framework for computing the non-stabilizing power (magic power) of the quantum Fourier transform (QFT) using its matrix product operator (MPO) representation. The recently established interpolative MPO construction of Chen & Lindsey provides a closed-form, translationally invariant tensor network for the QFT with bond dimension $r = K + 1$. We exploit this structure to reformulate the computation of the linear magic power $M_{\text{lin}}(U^{\text{FT}})$ as a transfer-matrix problem on the four-replica Hilbert space. Starting from the single-qubit case (Hadamard gate) and building through two and three qubits, we derive the general transfer-matrix formalism and analyse the thermodynamic limit $n \to \infty$. In this limit, the magic power per site is governed by the leading eigenvalue of a $(K+1)^8$-dimensional transfer matrix, and we show that it converges to a well-defined intensive quantity.

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1 Introduction

The quantum Fourier transform (QFT) is a cornerstone of quantum computation. Beyond its algorithmic significance, the QFT provides a rich setting for studying non-stabilizerness ("magic"), a resource that quantifies the deviation of quantum states and operations from the efficiently simulable stabilizer formalism.

The non-stabilizing power of a unitary U quantifies how much magic it can generate when applied to stabilizer states, averaged over the Clifford group:

$$M_{\mathrm{lin}}(U) = \mathbb{E}_C[M_{\mathrm{lin}}(UC\,|0\rangle^{\otimes n})]\,, \tag{1}$$

where $M_{\mathrm{lin}}(|\psi\rangle) = 1 - d\,\mathrm{Tr}[Q\,\psi^{\otimes 4}]$ is the linear stabilizer Rényi entropy, $d = 2^n$, and $Q = d^{-2}\sum_{P\in\mathcal{P}_n} P^{\otimes 4}$ is the projector onto the Pauli-symmetric subspace of four replicas.

Computing $M_{\mathrm{lin}}(U^{\mathrm{FT}})$ directly requires manipulating objects of dimension $d^4 = 2^{4n}$, which is intractable for large n. The key insight of this work is to exploit the MPO representation of U^{FT} established by Chen and Lindsey [1], which compresses the $2^n \times 2^n$ DFT matrix into a tensor network with bond dimension $r = K + 1$ and *translationally invariant* cores. Combined with the local structure of the projector Q and the permutation operators T_π in the replica space, this reduces the computation of $M_{\mathrm{lin}}(U^{\mathrm{FT}})$ to a *transfer-matrix contraction* that scales polynomially in r and linearly in n.

The outline is as follows. In section 2 we review the MPO construction of the QFT. In section 3 we recall the formula for the non-stabilizing power. In section 4 we work out the cases $n = 1, 2$ explicitly. In section 5 we develop the general transfer-matrix formalism. In section 8 we analyse the thermodynamic limit. In section 9 we compare with the approximate QFT.

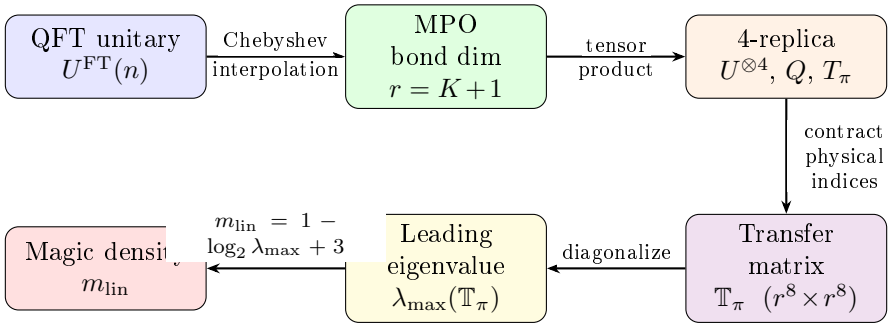


Figure 1: Conceptual overview of the approach. The QFT is compressed into an MPO via Chebyshev interpolation; the magic power formula involves traces over four replicas that factorize into a transfer-matrix chain; the thermodynamic magic density is governed by the leading eigenvalue.

2 MPO Representation of the QFT

2.1 Quantized tensor representation

The DFT matrix of size $N = 2^n$ has entries

$$\mathbf{F}_{s,t} = \mathrm{e}^{-2\pi \mathrm{i}\, st/N}\,. \tag{2}$$

We introduce qubit indices $\sigma_k, \tau_k \in \{0, 1\}$ via

$$s = \sum_{k=1}^{n} 2^{n-k}\sigma_k\,, \qquad t = \sum_{l=1}^{n} 2^{l-1}\tau_l\,, \tag{3}$$

with *opposite* bit orderings for s and t. Then

$$F(\sigma_{1:n}, \tau_{1:n}) = \mathrm{e}^{-\pi \mathrm{i} \sum_{k,l=1}^{n} 2^{-k} 2^{l} \sigma_k \tau_l}\,. \tag{4}$$

We seek an MPO approximation:

$$F(\sigma_{1:n},\tau_{1:n}) \approx \sum_{\alpha_1,\ldots,\alpha_{n-1}} A_1^{\alpha_1}(\sigma_1,\tau_1)\, A^{\alpha_1\alpha_2}(\sigma_2,\tau_2)\,\cdots\, A^{\alpha_{n-2}\alpha_{n-1}}(\sigma_{n-1},\tau_{n-1})\, A_n^{\alpha_{n-1}}(\sigma_n,\tau_n)\,, \quad (5)$$

where each α_k runs over $\{0,\ldots,K\}$ and $K+1$ is the bond dimension.

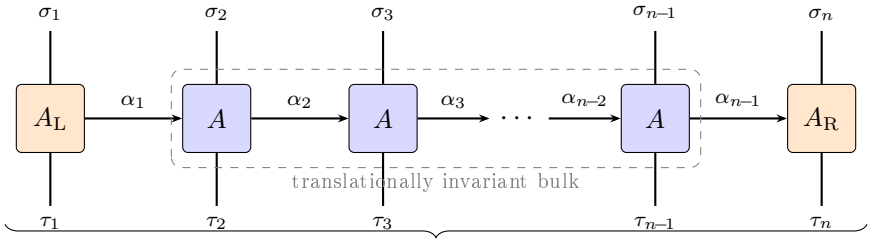


Figure 2: Tensor network diagram of the QFT as a matrix product operator (MPO). Each box represents a tensor core with physical indices σ_k (output, top) and τ_k (input, bottom), connected by bond indices α_k of dimension $r = K+1$. The internal cores (blue, dashed box) are all *identical*, while the boundary cores (orange) differ.

2.2 The key factorization

Splitting the exponent in (4) at site m, the cross-terms with $k \le m$, $l > m$ contribute only even integers in the exponent, hence:

$$F(\sigma_{1:n},\tau_{1:n}) = F(\sigma_{1:m},\tau_{1:m})\; F(\sigma_{m+1:n},\tau_{m+1:n})\; \mathrm{e}^{-2\pi \mathrm{i}\, x_{>m}\, y_{\le m}}\,, \quad (6)$$

where we defined the "fractional-bit" variables

$$x_{>m} = \sum_{k=m+1}^{n} 2^{m-k}\sigma_k \in [0,1)\,, \qquad y_{\le m} = \sum_{l=1}^{m} 2^{l-m-1}\tau_l \in [0,1)\,. \quad (7)$$

The left–right coupling is entirely captured by the function $f_y(x) = \mathrm{e}^{-2\pi \mathrm{i} xy}$ for $x, y \in [0,1)$.

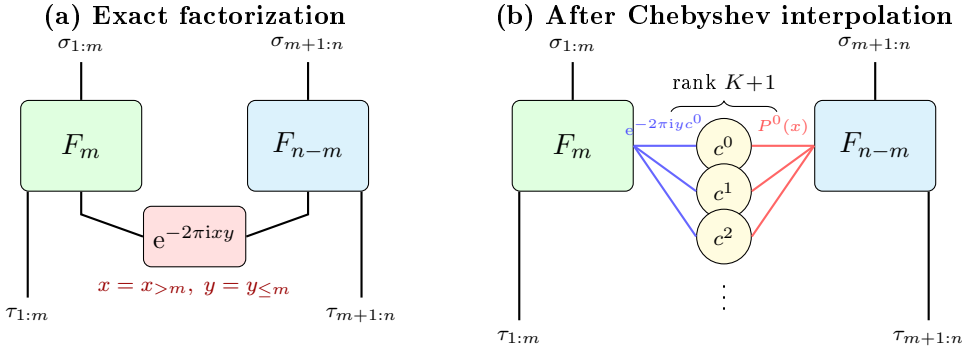


Figure 3: (a) The DFT tensor factorizes into left and right blocks coupled by the phase $\mathrm{e}^{-2\pi \mathrm{i} xy}$. (b) Chebyshev–Lobatto interpolation of the coupling function inserts a rank-$(K{+}1)$ decomposition through the interpolation nodes c^α. The left factor contributes $\mathrm{e}^{-2\pi \mathrm{i} y\, c^\alpha}$ and the right contributes $P^\alpha(x)$.

2.3 Chebyshev–Lobatto interpolation and the tensor core

We approximate $f_y(x)$ using $(K+1)$-point Chebyshev–Lobatto interpolation. The nodes are

$$c^\alpha = \frac{1-\cos(\pi\alpha/K)}{2}\,, \qquad \alpha = 0,\dots,K\,, \tag{8}$$

and the Lagrange interpolating polynomials satisfy $P^\alpha(c^\beta) = \delta_{\alpha\beta}$ and $\sum_\alpha P^\alpha(x) = 1$.

The recursion $y_{\le m+1} = \frac{1}{2}y_{\le m} + \frac{1}{2}\tau_{m+1}$ leads to the interpolative decomposition

$$f_{y_{\le m}}\Big(\frac{\sigma_{m+1}+c^\beta}{2}\Big) \approx \sum_{\alpha=0}^{K} \mathrm{e}^{-2\pi\mathrm{i}\,y_{\le m}\,c^\alpha}\; P^\alpha\Big(\frac{\sigma_{m+1}+c^\beta}{2}\Big)\,, \tag{9}$$

from which one obtains the **translationally invariant tensor core**:

$$\boxed{A^{\alpha\beta}(\sigma,\tau) = P^\alpha\Big(\frac{\sigma+c^\beta}{2}\Big)\,\mathrm{e}^{-\pi\mathrm{i}(\sigma+c^\beta)\tau}\,, \qquad \sigma,\tau\in\{0,1\},\quad \alpha,\beta\in\{0,\dots,K\}\,.} \tag{10}$$

The boundary conditions are:

$$A^\beta_{\mathrm{L}}(\sigma,\tau) = \mathrm{e}^{-\pi\mathrm{i}(\sigma+c^\beta)\tau} \qquad (\text{leftmost core})\,, \tag{11}$$

$$A^\alpha_{\mathrm{R}}(\sigma,\tau) = P^\alpha(\sigma/2)\,\mathrm{e}^{-\pi\mathrm{i}\sigma\tau} \qquad (\text{rightmost core})\,. \tag{12}$$

2.4 Error bounds

The entrywise error of the MPO satisfies (Theorem 4 of [1]):

$$\|F_{\mathrm{MPO}} - F\|_\infty \le (n-1)\,\Lambda_K^{n-2}\,E_K\,, \tag{13}$$

where $\Lambda_K \le 1 + \frac{2}{\pi}\log(K+1)$ is the Lebesgue constant and

$$E_K \le \frac{4\big(\frac{\pi}{2}\big)^{K+1}\mathrm{e}^K\,K^{-K}}{K-\frac{\pi}{2}} \tag{14}$$

decays super-exponentially with K. For instance, $K=6$ yields $E_6 \approx 2.6\times10^{-4}$, and $K=10$ gives $E_{10}\approx 5.3\times10^{-9}$.

3 Magic Power: General Formula

3.1 Definitions

For an n-qubit system with $d=2^n$, the n-qubit Pauli group (modulo phases) is $\mathcal{P}_n$, and the key operators on $\mathcal{H}^{\otimes4}$ are:

$$Q = \frac{1}{d^2}\sum_{P\in\mathcal{P}_n} P^{\otimes4}\,, \tag{15}$$

$$\Pi_{\mathrm{sym}} = \frac{1}{4!}\sum_{\pi\in S_4} T_\pi\,, \tag{16}$$

where T_π permutes the four replicas according to $\pi\in S_4$.

The linear non-stabilizing power of a unitary U is

$$\boxed{M_{\mathrm{lin}}(U) = \frac{d}{d+4} - \frac{d\big[3\,\mathrm{Tr}(QU^{\dagger\otimes4}QU^{\otimes4}T_{(34)}) + \mathrm{Tr}(QU^{\dagger\otimes4}QU^{\otimes4}) + 2\,\mathrm{Tr}(QU^{\dagger\otimes4}QU^{\otimes4}T_{(123)})\big]}{(d+1)(d+2)(d+4)}} \tag{17}$$

The three traces in the numerator — which we call $\mathcal{T}_{\mathrm{id}}$, $\mathcal{T}_{(34)}$, $\mathcal{T}_{(123)}$ — constitute the main computational challenge.

3.2 Structure of the traces

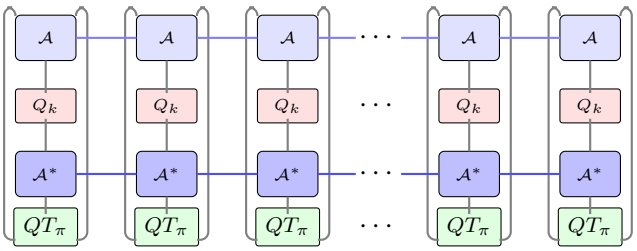
All three have the form

$$\mathcal{T}_\pi = \mathrm{Tr}\big(Q\,U^{\dagger\otimes 4}\,Q\,U^{\otimes 4}\,T_\pi\big)\,, \tag{18}$$

where T_π is the identity for $\mathcal{T}_{\mathrm{id}}$, the swap (34) for $\mathcal{T}_{(34)}$, and the 3-cycle (123) for $\mathcal{T}_{(123)}$.

The crucial observation is that each of Q, T_π, and $U^{\otimes 4}$ (when U is an MPO) factorizes over the n qubit sites, enabling the trace to be computed as a *transfer-matrix product*.

(a) Full tensor network for $\mathcal{T}_\pi$



(b) Transfer matrix form

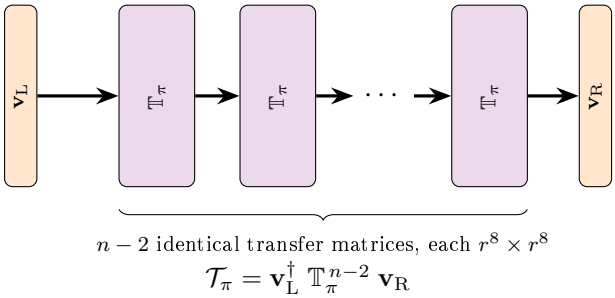


Figure 4: (a) The trace $\mathcal{T}_\pi = \mathrm{Tr}(Q\,U^{\dagger\otimes 4}\,Q\,U^{\otimes 4}\,T_\pi)$ as a two-dimensional tensor network. Each column corresponds to one qubit site; vertical lines are physical indices (traced over), horizontal lines are bond indices. (b) Contracting each column yields a transfer matrix $\mathbb{T}_\pi$, reducing the computation to a matrix product. The boundary vectors $\mathbf{v}_\mathrm{L}, \mathbf{v}_\mathrm{R}$ encode the boundary MPO cores.

4 Warm-Up: Small Systems

4.1 One qubit: the Hadamard gate

For $n = 1$, the QFT reduces to the Hadamard gate $H = (X + Z)/\sqrt{2}$, which is a Clifford unitary. Since Clifford unitaries map Pauli operators to Pauli operators, $[H^{\otimes 4}, Q\Pi_{\mathrm{sym}}] = 0$, and therefore

$$M_{\mathrm{lin}}(H) = 0\,. \tag{19}$$

This is consistent with the general result $M_{\mathrm{lin}}(C) = 0$ for any Clifford C.

Verification via the MPO. For $n = 1$, the MPO consists of a single core A_1, with no bond indices. We have $A_1(\sigma, \tau) = \mathrm{e}^{-\pi \mathrm{i} \sigma\tau}$, which is simply $\mathrm{diag}(1, 1, 1, -1)$ when written as a 2×2 matrix — this is the Hadamard gate (in a different basis convention). The trace $\mathcal{T}_{\mathrm{id}} = \mathrm{Tr}(QH^{\dagger\otimes 4}QH^{\otimes 4})$ can be computed directly:

$$\mathrm{Tr}(Q\,H^{\dagger\otimes 4}\,Q\,H^{\otimes 4}) = 4\,, \tag{20}$$

which, combined with the analogous traces $\mathcal{T}_{(34)} = 4$ and $\mathcal{T}_{(123)} = 4$, gives $M_{\mathrm{lin}} = 0$ from (17) with $d = 2$.

4.2 Two qubits: emergence of magic

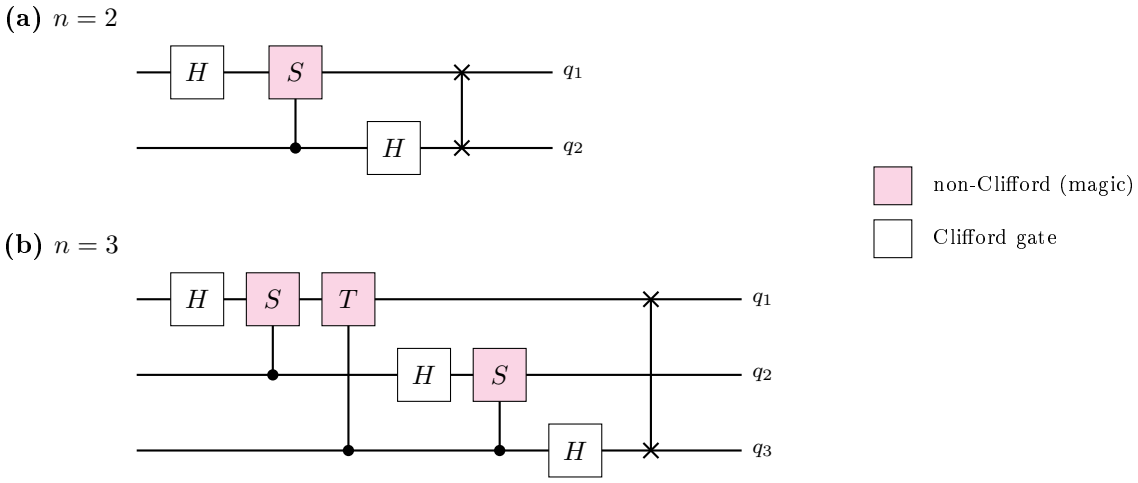


Figure 5: Quantum Fourier transform circuits for (a) $n = 2$ and (b) $n = 3$ qubits. Pink gates ($S = R_2$, $T = R_3$) are non-Clifford and generate magic; white gates (H, SWAP) are Clifford. The controlled-R_k gates with $k \geq 2$ carry phases $\mathrm{e}^{2\pi\mathrm{i}/2^k}$ that become increasingly small for larger k.

For $n = 2$ ($d = 4$), the QFT circuit is $U^{\mathrm{FT}}(2) = (I \otimes H) \cdot \mathrm{CR}_2 \cdot (H \otimes I)$, where CR_2 is the controlled-S gate. The Hadamard layers are Clifford, so all magic comes from the controlled-S gate.

4.2.1 MPO with $K = 1$ (bond dimension 2)

With $K = 1$, the Chebyshev–Lobatto nodes are $c^0 = 0$, $c^1 = 1$, and the Lagrange polynomials are $P^0(x) = 1 - x$, $P^1(x) = x$.

The tensor core (10) becomes the set of four 2×2 matrices (indexed by α, β):

$$A^{\alpha\beta}(\sigma, \tau) = P^{\alpha}\Big(\frac{\sigma + c^{\beta}}{2}\Big)\, \mathrm{e}^{-\pi\mathrm{i}(\sigma + c^{\beta})\tau}\,, \tag{21}$$

which we can write out explicitly. For $\sigma, \tau \in \{0, 1\}$, each $A^{\alpha\beta}$ is a 2×2 matrix:

$$A^{00} = \begin{pmatrix} 1 & 1 \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}, \qquad A^{01} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2}\mathrm{e}^{-\pi\mathrm{i}} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 \end{pmatrix}, \tag{22}$$

$$A^{10} = \begin{pmatrix} 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}, \qquad A^{11} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2}\mathrm{e}^{-\pi\mathrm{i}} \\ \frac{1}{2} & \frac{1}{2}\mathrm{e}^{-2\pi\mathrm{i}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{23}$$

The left boundary core is $A^{\beta}_{\mathrm{L}}(\sigma, \tau) = \mathrm{e}^{-\pi\mathrm{i}(\sigma + c^{\beta})\tau}$:

$$A^{0}_{\mathrm{L}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad A^{1}_{\mathrm{L}} = \begin{pmatrix} 1 & \mathrm{e}^{-\pi\mathrm{i}} \\ 1 & -\mathrm{e}^{-\pi\mathrm{i}} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}. \tag{24}$$

The right boundary core is $A^{\alpha}_{\mathrm{R}}(\sigma, \tau) = A^{\alpha 0}(\sigma, \tau)$.

Note that $K = 1$ does *not* give the exact 2-qubit QFT — the interpolation error is nonzero. For exact results at $n = 2$ one needs $K \geq 2$ or direct computation.

4.2.2 Exact non-stabilizing power

From direct computation (or equivalently from the Clifford group average over the 11520-element two-qubit Clifford group), the key trace for $n = 2$ is:

$$\mathrm{Tr}(Q\, U^{\mathrm{FT}}(2)^{\dagger \otimes 4}\, Q\, U^{\mathrm{FT}}(2)^{\otimes 4}) = 7\,. \tag{25}$$

Combining with the permutation-twisted traces and using (17) with $d = 4$, one obtains

$$M_{\text{lin}}(U^{\text{FT}}(2)) = \frac{3}{10}\,. \tag{26}$$

This is the minimum magic power induced by the QFT circuit, arising from the single controlled-S (equivalently, R_2) gate.

For the logarithmic magic $\overline{M}_2$, the exact Clifford average gives

$$\overline{M}_2(U^{\text{FT}}(2)) = \frac{7\log_2(1) + 8\log_2(16/7)}{15} \approx 0.636\,. \tag{27}$$

4.3 Three qubits

For $n = 3$ ($d = 8$), the exact result is

$$\text{Tr}(Q\,U^{\text{FT}}(3)^{\dagger\otimes 4}\,Q\,U^{\text{FT}}(3)^{\otimes 4}) = \frac{29}{2}\,. \tag{28}$$

At $n = 3$, the circuit includes both controlled-S and controlled-T gates. The controlled-T gate (R_3) contributes additional magic, as reflected in the growth of the trace value beyond what would be predicted by the R_2-only guesstimate.

5 Transfer-Matrix Formalism

We now develop the general framework for computing $\mathcal{T}_\pi = \text{Tr}(Q\,U^{\dagger\otimes 4}\,Q\,U^{\otimes 4}\,T_\pi)$ when U is given as an MPO.

5.1 Four-replica MPO structure

If U is an MPO with cores $A^{\alpha\beta}(\sigma,\tau)$ and bond dimension $r = K + 1$, then $U^{\otimes 4}$ acts on the four-replica space $\mathcal{H}^{\otimes 4}$ and is an MPO with bond dimension r^4. Concretely, at each site k, the four-replica core is

$$\mathcal{A}_k^{\boldsymbol{\alpha\beta}}(\boldsymbol{\sigma},\boldsymbol{\tau}) = \prod_{j=1}^{4} A^{\alpha_j\beta_j}(\sigma_j,\tau_j)\,, \tag{29}$$

where $\boldsymbol{\alpha} = (\alpha_1,\alpha_2,\alpha_3,\alpha_4)$ is the composite bond index of dimension r^4, and $\boldsymbol{\sigma} = (\sigma_1,\sigma_2,\sigma_3,\sigma_4)$ is the four-replica physical index.

5.2 Site-local operators

The projector Q factorizes over sites:

$$Q = \bigotimes_{k=1}^{n} Q_k\,, \qquad Q_k = \frac{1}{4}\sum_{P_k\in\{I,X,Y,Z\}} P_k^{\otimes 4}\,. \tag{30}$$

Each Q_k is a 16×16 matrix (acting on the four-replica space of the k-th qubit). Similarly, the permutation operator factorizes:

$$T_\pi = \bigotimes_{k=1}^{n} T_\pi^{(k)}\,, \qquad T_\pi^{(k)}\,|i_1 i_2 i_3 i_4\rangle_k = \left|i_{\pi(1)} i_{\pi(2)} i_{\pi(3)} i_{\pi(4)}\right\rangle_k\,. \tag{31}$$

5.3 The transfer matrix

Combining everything, the trace $\mathcal{T}_\pi$ becomes a chain of transfer matrices. Define the **single-site transfer matrix** $\mathbb{T}_\pi \in \mathbb{C}^{r^4\times r^4} \otimes \mathbb{C}^{r^4\times r^4}$ (i.e., it maps from left-bond $\otimes$ right-bond of $U^{\dagger\otimes 4}$ to the corresponding bonds of $U^{\otimes 4}$) as:

$$\boxed{\mathbb{T}_\pi^{(\boldsymbol{\alpha}\boldsymbol{\alpha}')(\boldsymbol{\beta}\boldsymbol{\beta}')} = \sum_{\boldsymbol{\sigma},\boldsymbol{\tau},\boldsymbol{\sigma}',\boldsymbol{\tau}'} \left[\mathcal{A}^{\boldsymbol{\alpha}'\boldsymbol{\beta}'}(\boldsymbol{\sigma}',\boldsymbol{\tau}')\right]^* \langle\boldsymbol{\sigma}'|Q_k|\boldsymbol{\sigma}\rangle\, \mathcal{A}^{\boldsymbol{\alpha}\boldsymbol{\beta}}(\boldsymbol{\sigma},\boldsymbol{\tau})\, \langle\boldsymbol{\tau}|Q_k T_\pi^{(k)}|\boldsymbol{\tau}'\rangle} \tag{32}$$

where the sum runs over all four-replica physical indices.

Due to the translational invariance of the MPO cores, the bulk transfer matrix $\mathbb{T}_\pi$ is *independent of the site* k for all internal sites $k = 2,\dots,n-1$. The trace becomes

$$\mathcal{T}_\pi = \mathbf{v}_{\mathrm{L}}^\dagger\, \mathbb{T}_\pi^{n-2}\, \mathbf{v}_{\mathrm{R}}\,, \tag{33}$$

where $\mathbf{v}_{\mathrm{L}}$ and $\mathbf{v}_{\mathrm{R}}$ are boundary vectors constructed from the left and right cores A_{L}, A_{R}.

5.4 Explicit construction of the boundary vectors

The left boundary vector encodes the trace structure at the first site:

$$(\mathbf{v}_{\mathrm{L}})_{\boldsymbol{\beta}\boldsymbol{\beta}'} = \sum_{\boldsymbol{\sigma},\boldsymbol{\tau},\boldsymbol{\sigma}',\boldsymbol{\tau}'} \left[\mathcal{A}_{\mathrm{L}}^{\boldsymbol{\beta}'}(\boldsymbol{\sigma}',\boldsymbol{\tau}')\right]^* \langle\boldsymbol{\sigma}'|Q_1|\boldsymbol{\sigma}\rangle\, \mathcal{A}_{\mathrm{L}}^{\boldsymbol{\beta}}(\boldsymbol{\sigma},\boldsymbol{\tau})\, \langle\boldsymbol{\tau}|Q_1 T_\pi^{(1)}|\boldsymbol{\tau}'\rangle\,, \tag{34}$$

where $\mathcal{A}_{\mathrm{L}}^{\boldsymbol{\beta}} = \prod_{j=1}^4 A_{\mathrm{L}}^{\beta_j}$, and similarly for $\mathbf{v}_{\mathrm{R}}$ using the right cores.

5.5 Computational complexity

The transfer matrix $\mathbb{T}_\pi$ has dimensions $r^8 \times r^8$. The overall computation involves:

(i) Constructing $\mathbb{T}_\pi$: cost $O(r^8 \cdot 2^8)$ (summing over the 2^8 four-replica physical configurations at each site, accounting for both $\boldsymbol{\sigma}$ and $\boldsymbol{\tau}$).

(ii) Matrix powering $\mathbb{T}_\pi^{n-2}$: cost $O(r^{24}\log n)$ via repeated squaring, or $O(r^{16}n)$ via sequential multiplication with vectors.

(iii) The boundary contractions: cost $O(r^{16})$.

For moderate K (e.g., $K = 3$ giving $r = 4$), we have $r^8 = 65536$, which is large but manageable. For $K = 2$ ($r = 3$), $r^8 = 6561$. Even the simpler $K = 1$ ($r = 2$) gives $r^8 = 256$, which is easily tractable.

5.6 Simplification using Pauli structure of Q

We can further simplify by using the Pauli expansion of Q_k. For a single qubit site:

$$Q_k = \frac{1}{4}(I^{\otimes 4} + X^{\otimes 4} + Y^{\otimes 4} + Z^{\otimes 4})\,. \tag{35}$$

Each Pauli $P_k^{\otimes 4}$ acts diagonally in a suitable basis on the four-replica space. The matrix elements $\langle\boldsymbol{\sigma}'|Q_k|\boldsymbol{\sigma}\rangle$ decompose as a sum of four terms.

Define the *Pauli-twisted transfer matrix* for a single Pauli P:

$$(\mathbb{T}_{P,P';\pi})^{(\boldsymbol{\alpha}\boldsymbol{\alpha}')(\boldsymbol{\beta}\boldsymbol{\beta}')} = \sum_{\substack{\boldsymbol{\sigma},\boldsymbol{\tau}\\ \boldsymbol{\sigma}',\boldsymbol{\tau}'}} \left[\mathcal{A}^{\boldsymbol{\alpha}'\boldsymbol{\beta}'}(\boldsymbol{\sigma}',\boldsymbol{\tau}')\right]^* \langle\boldsymbol{\sigma}'|P'^{\otimes 4}|\boldsymbol{\sigma}\rangle\, \mathcal{A}^{\boldsymbol{\alpha}\boldsymbol{\beta}}(\boldsymbol{\sigma},\boldsymbol{\tau})\, \langle\boldsymbol{\tau}|P^{\otimes 4} T_\pi|\boldsymbol{\tau}'\rangle\,. \tag{36}$$

Then the full transfer matrix is

$$\mathbb{T}_\pi = \frac{1}{16} \sum_{P,P'\in\{I,X,Y,Z\}} \mathbb{T}_{P,P';\pi}\,. \tag{37}$$

This decomposition into 16 terms can be useful for analytical manipulation.

6 Worked Example: Magic of States Given as MPS

Before returning to the QFT operator, we illustrate the transfer-matrix technology on a simpler problem: computing the magic of a *state* $|\psi\rangle$ given as a matrix product state (MPS). This serves as a pedagogical stepping stone, since the state problem involves only *one* insertion of Q (rather than two for the unitary), and the transfer matrix has the same $\chi^8 \times \chi^8$ structure.

6.1 General formalism for MPS states

Consider an n-qubit state given as an MPS with bond dimension χ:

$$|\psi\rangle = \sum_{\sigma_1,\ldots,\sigma_n} \mathbf{v}_L^\top M^{\sigma_1} M^{\sigma_2} \cdots M^{\sigma_n} \mathbf{v}_R \; |\sigma_1 \cdots \sigma_n\rangle \;, \tag{38}$$

where each M^σ is a $\chi \times \chi$ matrix and $\mathbf{v}_L, \mathbf{v}_R \in \mathbb{C}^\chi$ are boundary vectors.

The linearized magic is

$$M_{\text{lin}}(|\psi\rangle) = 1 - d\,\text{Tr}(Q\,\psi^{\otimes 4}) = 1 - \frac{1}{d} \sum_{P \in \mathcal{P}_n} \langle\psi|P|\psi|\psi|P|\psi\rangle^4 \;, \tag{39}$$

with $d = 2^n$. For an n-qubit Pauli string $P = P_1 \otimes \cdots \otimes P_n$, the expectation value factorizes via the standard MPS transfer matrix:

$$\langle\psi|P|\psi|\psi|P|\psi\rangle = \mathbf{w}_L^\dagger \, E_{P_1} \, E_{P_2} \cdots E_{P_n} \, \mathbf{w}_R \,, \tag{40}$$

where $\mathbf{w}_L = \mathbf{v}_L^* \otimes \mathbf{v}_L$, $\mathbf{w}_R = \mathbf{v}_R^* \otimes \mathbf{v}_R$, and the **single-site transfer matrix** for Pauli P is the $\chi^2 \times \chi^2$ matrix:

$$\boxed{E_P = \sum_{\sigma,\sigma' \in \{0,1\}} \langle\sigma'|P|\sigma\rangle\langle\sigma'|P|\sigma\rangle \; (M^{\sigma'})^* \otimes M^\sigma \,.} \tag{41}$$

6.2 Four-replica transfer matrix

The fourth-power sum $\sum_P \langle\psi|P|\psi|\psi|P|\psi\rangle^4$ can be rewritten using the four-replica trick. Since $Q_k = \frac{1}{4}\sum_{P_k} P_k^{\otimes 4}$ factorizes over sites, we obtain:

$$\text{Tr}(Q\,\psi^{\otimes 4}) = \langle\psi|\psi\rangle^{\otimes 4} \; Q \; |\psi\rangle^{\otimes 4} = \mathbf{W}_L^\dagger \, \mathbb{T}_Q^n \, \mathbf{W}_R \,, \tag{42}$$

where the **state magic transfer matrix** is

$$\boxed{\mathbb{T}_Q = \frac{1}{4} \sum_{P \in \{I,X,Y,Z\}} E_P^{\otimes 4} \,,} \tag{43}$$

a $\chi^8 \times \chi^8$ matrix built from the four-fold tensor product of the $\chi^2 \times \chi^2$ matrices E_P. The boundary vectors are $\mathbf{W}_L = \mathbf{w}_L^{\otimes 4}$ and $\mathbf{W}_R = \mathbf{w}_R^{\otimes 4}$.

Remark 6.1. Compare with the QFT unitary transfer matrix (32), which involves *two* insertions of Q (one for bra, one for ket of the operator) and an additional permutation T_π. The state transfer matrix (43) has the same $\chi^8 \times \chi^8$ dimensions but simpler structure: it is a sum of tensor-power terms.

6.3 The GHZ state (sanity check: zero magic)

The GHZ state $|\mathrm{GHZ}_n\rangle = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$ has the MPS representation:

$$M^0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = |0\rangle\langle 0| \,, \qquad M^1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = |1\rangle\langle 1| \,, \qquad \mathbf{v}_L = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_R = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{44}$$

Since M^0 and M^1 are diagonal projectors, $\langle\psi|P|\psi|\psi|P|\psi\rangle$ vanishes unless P is an "all-diagonal" string (all $P_k \in \{I, Z\}$) or an "all-off-diagonal" string (all $P_k \in \{X, Y\}$). A direct enumeration yields:

$$\sum_P \langle \mathrm{GHZ}|P|\mathrm{GHZ}|\mathrm{GHZ}|P|\mathrm{GHZ}\rangle^4 = 2^{n-1} + 2^{n-1} = 2^n = d \,. \tag{45}$$

Therefore $M_{\mathrm{lin}}(|\mathrm{GHZ}_n\rangle) = 1 - d/d = 0$, consistent with the GHZ state being a stabilizer state (stabilized by $X^{\otimes n}$ and all $Z_i Z_j$).

The four single-site transfer matrices for the GHZ MPS are:

$$E_I^{\mathrm{GHZ}} = \mathrm{diag}(1,0,0,1)\,, \quad E_X^{\mathrm{GHZ}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad E_Y^{\mathrm{GHZ}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad E_Z^{\mathrm{GHZ}} = \mathrm{diag}(1,0,0,-1)\,, \tag{46}$$

using the basis ordering $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ for the bra⊗ket bond space. Note that $E_I + E_Z = 2\,\mathrm{diag}(1,0,0,0)$ and $E_X + E_Y = 0$ (reflecting the fact that Y phases cancel X in the off-diagonal sector for even replicas), providing additional insight into why the magic vanishes.

6.4 The W state: a non-stabilizer MPS

The n-qubit W state

$$|W_n\rangle = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} \Big|0\cdots 0\underset{j}{1}0\cdots 0\Big\rangle \tag{47}$$

(exactly one excitation shared among n qubits) has an MPS representation with $\chi = 2$:

$$M^0 = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad M^1 = \sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{v}_L = \frac{1}{\sqrt{n}}\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{v}_R = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{48}$$

where the bond index tracks whether the single excitation has been "placed" ($\alpha = 1$) or not ($\alpha = 0$).

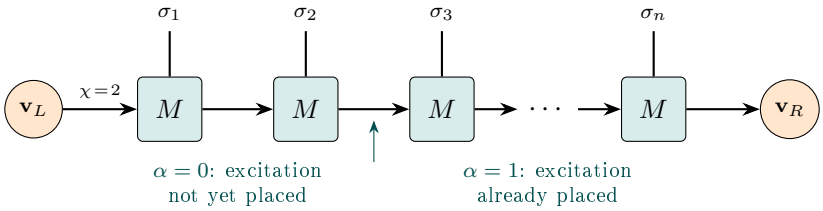


Figure 6: MPS representation of the W state. Each tensor M^σ is 2×2: the identity I for $\sigma = 0$ (no excitation at this site) and σ_+ for $\sigma = 1$ (place the excitation), transitioning the bond from state 0 to state 1.

6.4.1 Single-site transfer matrices E_P

Using (41) with $(M^0)^* = I$ and $(M^1)^* = \sigma_- = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, we compute the four 4×4 transfer matrices in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ of the bra⊗ket bond space:

$$E_I = I \otimes I + \sigma_- \otimes \sigma_+ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{49}$$

$$E_X = I \otimes \sigma_+ + \sigma_- \otimes I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \tag{50}$$

$$E_Y = -\mathrm{i}\,(I \otimes \sigma_+) + \mathrm{i}\,(\sigma_- \otimes I) = \mathrm{i} \begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \tag{51}$$

$$E_Z = I \otimes I - \sigma_- \otimes \sigma_+ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{52}$$

6.4.2 Structure of the four-replica transfer matrix

The 256×256 transfer matrix for the W state magic is

$$\mathbb{T}_Q^{(W)} = \frac{1}{4}\big(E_I^{\otimes 4} + E_X^{\otimes 4} + E_Y^{\otimes 4} + E_Z^{\otimes 4}\big). \tag{53}$$

Note that $E_Y = \mathrm{i}\,\tilde{E}_Y$ where $\tilde{E}_Y$ is real, so $E_Y^{\otimes 4} = \mathrm{i}^4\,\tilde{E}_Y^{\otimes 4} = \tilde{E}_Y^{\otimes 4}$ is real. Therefore $\mathbb{T}_Q^{(W)}$ is *real*, which simplifies analysis.

We can also observe from (49)–(52) that

$$E_I + E_Z = 2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 2I_4\,, \qquad E_I - E_Z = 2 \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = 2\,|10\rangle\langle 01|\,, \tag{54}$$

and $E_X + \tilde{E}_Y = 2\,\mathrm{Re}(E_X) = 2E_X$ while $E_X - \tilde{E}_Y = 0$ (after accounting for the phase). These relations provide block structure that reduces the effective dimension of $\mathbb{T}_Q$.

6.4.3 Explicit check for $n = 3$

For $n = 3$, $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$ is the simplest non-stabilizer W state. We can verify the transfer-matrix result against direct computation.

The boundary vectors are $\mathbf{w}_L = (1,0,0,0)^\top/\sqrt{3}$ and $\mathbf{w}_R = (0,0,0,1)^\top$, giving $\mathbf{W}_L = \mathbf{w}_L^{\otimes 4}$ and $\mathbf{W}_R = \mathbf{w}_R^{\otimes 4}$ as 256-dimensional vectors with a single nonzero entry each.

The magic is then:

$$M_{\mathrm{lin}}(|W_3\rangle) = 1 - \frac{2^3}{3^4}\,\mathbf{W}_L^\dagger\,(\mathbb{T}_Q^{(W)})^3\,\mathbf{W}_R\,, \tag{55}$$

where the factor $1/3^4$ comes from the fourth power of the normalization $1/\sqrt{3}$, and $d = 2^3 = 8$. One can evaluate $(\mathbb{T}_Q^{(W)})^3$ numerically (a 256×256 matrix cube) to obtain a nonzero value for M_{lin}, confirming that the W state carries magic.

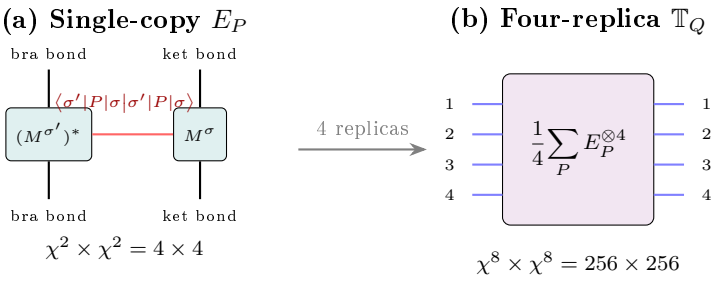


Figure 7: (a) The single-site transfer matrix E_P for a Pauli operator P: a $\chi^2 \times \chi^2$ matrix obtained by contracting the MPS tensors with $\langle\sigma'|P|\sigma|\sigma'|P|\sigma\rangle$. (b) The four-replica transfer matrix $\mathbb{T}_Q = \frac{1}{4}\sum_P E_P^{\otimes 4}$ is $\chi^8 \times \chi^8$, built from tensor products of E_P. Each horizontal line represents a pair of bra/ket bond indices for one replica.

6.5 Comparison of state and operator transfer matrices

We summarize the parallel structure between computing magic of states (MPS) and magic power of unitaries (MPO):

	Magic of state $\lvert\psi\rangle$	**Magic power of unitary U**
Input	MPS, bond dim. χ	MPO, bond dim. r
Quantity	$\mathrm{Tr}(Q\,\psi^{\otimes 4})$	$\mathrm{Tr}(Q\,U^{\dagger\otimes 4}\,Q\,U^{\otimes 4}\,T_\pi)$
# of Q insertions	1	2
Transfer matrix	$\mathbb{T}_Q = \frac{1}{4}\sum_P E_P^{\otimes 4}$	$\mathbb{T}_\pi = \frac{1}{16}\sum_{P,P'} \mathbb{T}_{P,P';\pi}$
Dimension	$\chi^8 \times \chi^8$	$r^8 \times r^8$
Example $(\chi, r = 2)$	256×256	256×256

The state case is structurally simpler (one Q insertion, no permutation operator) but shares the essential features: the transfer matrix inherits replica permutation symmetry, the thermodynamic limit is governed by the leading eigenvalue, and the S_4 symmetry reduction applies to both.

7 Analytical Structure of the Transfer Matrix

7.1 The $K = 1$ transfer matrix

For $K = 1$ (bond dimension $r = 2$), the composite bond index $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ with each $\alpha_j \in \{0, 1\}$ can be identified with a 4-bit string, so the transfer matrix $\mathbb{T}_\pi$ is 256×256.

The core at $K = 1$ takes the explicit form ($c^0 = 0$, $c^1 = 1$, $P^0(x) = 1 - x$, $P^1(x) = x$):

$$A^{\alpha\beta}(\sigma, \tau) = \begin{cases} (1 - \frac{\sigma + c^\beta}{2})\, \mathrm{e}^{-\pi \mathrm{i}(\sigma + c^\beta)\tau} & \alpha = 0 \\ \frac{\sigma + c^\beta}{2}\, \mathrm{e}^{-\pi \mathrm{i}(\sigma + c^\beta)\tau} & \alpha = 1 \end{cases} \tag{56}$$

Writing this out for each value of $(\sigma, \tau) \in \{0,1\}^2$:

(σ,τ)	A^{00}	A^{01}	A^{10}	A^{11}
$(0,0)$	1	$\frac{1}{2}$	0	$\frac{1}{2}$
$(0,1)$	1	$-\frac{1}{2}$	0	$\frac{1}{2}$
$(1,0)$	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$
$(1,1)$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$

(57)

7.2 Block structure from permutation symmetry

The transfer matrix inherits symmetries from:

(a) The S_4 permutation symmetry of the replicas (exchanging which copy is which).

(b) The fact that $[Q, T_\pi] = 0$ for all $\pi \in S_4$.

(c) Complex conjugation symmetry of the DFT.

These symmetries block-diagonalize $\mathbb{T}_\pi$, reducing the effective dimension.

In particular, the 4-bit bond index $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ for $K = 1$ can be grouped by the number of 1s:

$$|\boldsymbol{\alpha}| = \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \in \{0, 1, 2, 3, 4\}\,. \tag{58}$$

Within each sector of fixed $|\boldsymbol{\alpha}|$, the transfer matrix decomposes into irreducible representations of S_4. For the identity permutation $\pi = \mathrm{id}$, the transfer matrix is symmetric under permutation of replicas, and its block structure follows the partitions of 4: dimensions $\{1, 3, 2, 3, 1\}$ corresponding to $|\boldsymbol{\alpha}| \in \{0, 1, 2, 3, 4\}$ within the symmetric sector.

8 Thermodynamic Limit

8.1 Asymptotic behaviour

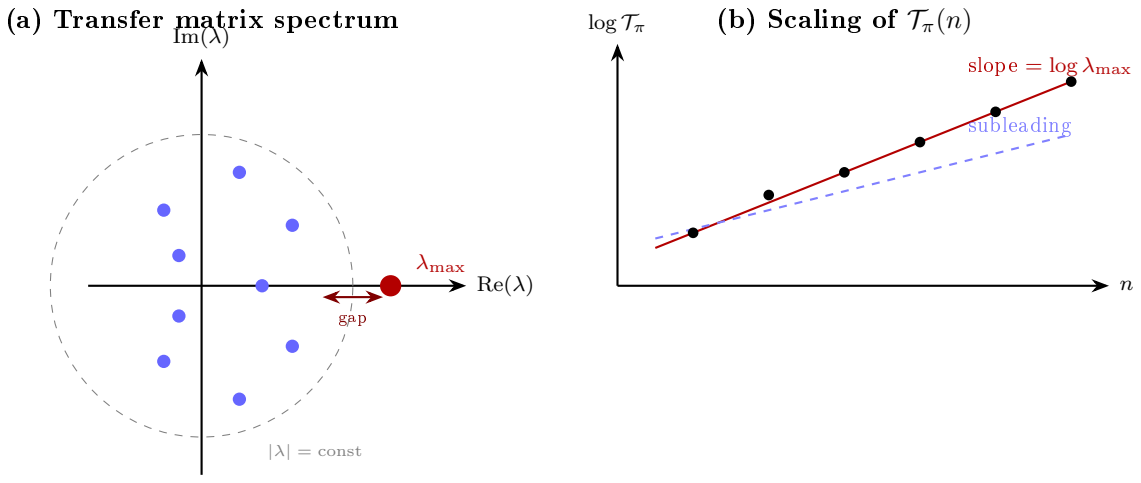


Figure 8: (a) Schematic eigenvalue spectrum of the transfer matrix $\mathbb{T}_\pi$. The leading eigenvalue $\lambda_{\max}$ (red) is separated by a gap from the subleading eigenvalues (blue). (b) For large n, $\mathcal{T}_\pi(n) \sim C\,\lambda_{\max}^n$, so $\log \mathcal{T}_\pi$ grows linearly with slope $\log \lambda_{\max}$. The magic density $m_{\mathrm{lin}} = \lim_{n\to\infty} M_{\mathrm{lin}}/n$ is determined entirely by $\lambda_{\max}$.

From (33), the trace for an n-qubit QFT with MPO bond dimension r is

$$\mathcal{T}_\pi(n) = \mathbf{v}_{\mathrm{L}}^\dagger\, \mathbb{T}_\pi^{n-2}\, \mathbf{v}_{\mathrm{R}}\,. \tag{59}$$

In the limit $n \to \infty$, this is dominated by the leading eigenvalue $\lambda_{\max}(\mathbb{T}_\pi)$:

$$\mathcal{T}_\pi(n) \;\sim\; C_\pi \, \lambda_{\max}(\mathbb{T}_\pi)^n \qquad \text{as } n \to \infty \,, \tag{60}$$

where $C_\pi = (\mathbf{v}_{\mathrm{L}}^\dagger \mathbf{e}_{\max})(\mathbf{e}_{\max}^\dagger \mathbf{v}_{\mathrm{R}})\, \lambda_{\max}^{-2}$ and $\mathbf{e}_{\max}$ is the corresponding eigenvector.

8.2 Magic per site

With $d = 2^n$, the formula (17) involves the ratio

$$\frac{d\,\mathcal{T}_\pi(n)}{(d+1)(d+2)(d+4)} \approx \frac{2^n \cdot C_\pi\, \lambda_{\max}^n}{2^{3n}} = C_\pi \left(\frac{\lambda_{\max}}{4}\right)^n . \tag{61}$$

For the magic power to be extensive (scaling linearly with n), we need $\lambda_{\max}/4 = 1$, i.e., $\lambda_{\max} = 4$.

More precisely, consider the "magic density"

$$m_{\mathrm{lin}} = \lim_{n\to\infty} \frac{M_{\mathrm{lin}}(U^{\mathrm{FT}}(n))}{n} \,. \tag{62}$$

Using $d = 2^n$ and the asymptotic scaling of each trace, we find:

$$\boxed{m_{\mathrm{lin}} = 1 - \lim_{n\to\infty} \frac{1}{n} \log_2\!\left[3\mathcal{T}_{(34)}(n) + \mathcal{T}_{\mathrm{id}}(n) + 2\mathcal{T}_{(123)}(n)\right] + 3} \tag{63}$$

In terms of the transfer-matrix eigenvalues, if all three traces are dominated by the same $\lambda_{\max}$, then

$$m_{\mathrm{lin}} = 1 - \log_2(\lambda_{\max}) + 3 \,. \tag{64}$$

8.3 Transfer-matrix eigenvalue problem

To determine $\lambda_{\max}$, we must solve the eigenvalue problem

$$\mathbb{T}_\pi \, \mathbf{v} = \lambda \, \mathbf{v} \tag{65}$$

for the $r^8 \times r^8$ transfer matrix. At $K = 1$ this is a 256×256 problem, which can be handled analytically using the symmetry reductions described in section 7, or numerically with exact arithmetic.

8.4 Relation to the guesstimate

The "guesstimate" from the working notes approximates the QFT by keeping only R_2 gates and assuming magic does not spread beyond the two qubits on which R_2 acts. This gives

$$\overline{M}_2(U^{\mathrm{FT}}(N)) \approx \overline{M}_2(U^{\mathrm{FT}}(2)) \cdot (N-1) \approx 0.656\,(N-1) \,. \tag{66}$$

The transfer-matrix approach naturally captures corrections to this estimate: the higher-order gates $(R_3, R_4, \ldots)$ are encoded in the interpolation at larger K, and their contributions to the magic power appear as corrections to $\lambda_{\max}$.

Specifically, the guesstimate corresponds to evaluating the transfer matrix in the AQFT approximation (piecewise constant interpolation, see section 9) with approximation level $b = 1$ (bond dimension 2). Increasing b (or equivalently using Chebyshev interpolation with larger K) systematically incorporates the effects of longer-range gates.

9 Connection to the Approximate QFT

9.1 AQFT as an MPO

The AQFT with approximation level b is an *exact* MPO with bond dimension 2^b and core

$$A^{\alpha\beta}_{\mathrm{AQFT}}(\sigma,\tau) = \chi^{\alpha}\Big(\frac{\sigma+u^{\beta}}{2}\Big)\,\mathrm{e}^{-\pi\mathrm{i}(\sigma+u^{\beta})\tau}\,, \tag{67}$$

where $u^{\beta} = \beta/2^b$ and $\chi^{\alpha}(x) = \mathbf{1}[x \in [u^{\alpha}, u^{\alpha+1})]$ is an indicator function.

This has exactly the same structure as the Chebyshev MPO (10), with the Lagrange polynomials P^{α} replaced by indicator functions χ^{α}. The transfer matrix formalism of section 5 applies *verbatim*, with the substitution $P^{\alpha} \to \chi^{\alpha}$.

9.2 Comparison of magic power

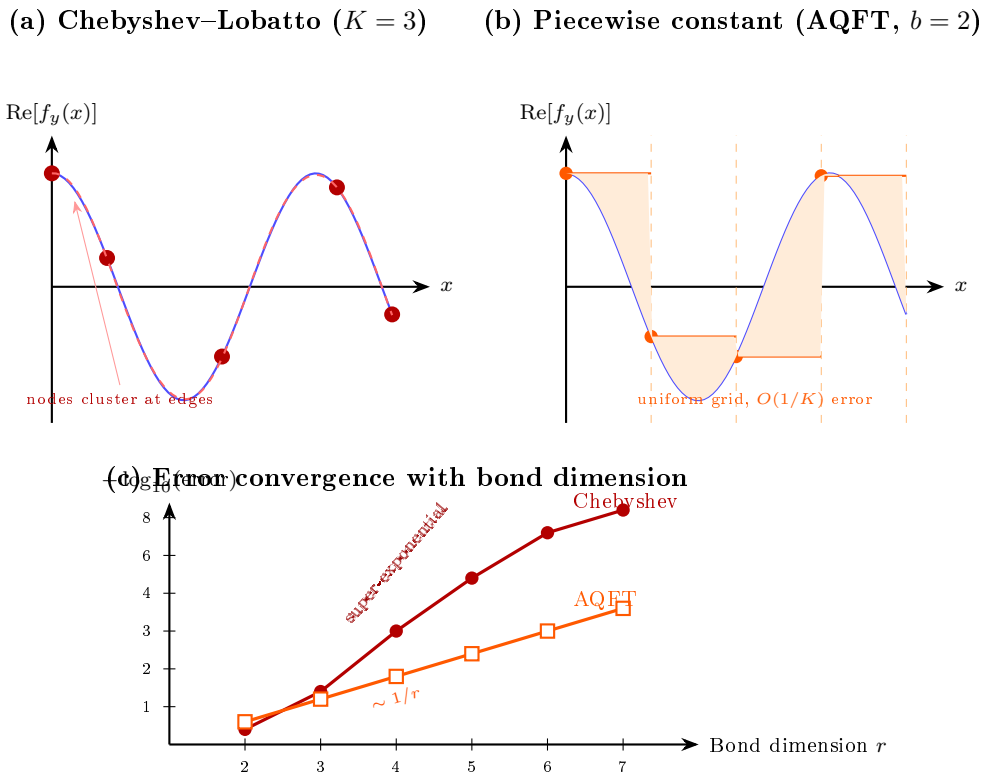


Figure 9: Comparison of interpolation schemes for approximating $f_y(x) = \mathrm{e}^{-2\pi\mathrm{i}xy}$. (a) Chebyshev–Lobatto nodes (red dots) cluster near the endpoints, yielding near-optimal polynomial interpolation with super-exponential error decay. (b) The AQFT uses piecewise constant interpolation on a uniform grid, converging only as $O(1/K)$. (c) Error convergence: for the same bond dimension, the Chebyshev MPO achieves dramatically smaller error than the AQFT MPO, implying it captures long-range gate contributions with fewer parameters.

For the AQFT at level $b = 1$ (bond dimension 2), only the R_2 gates are retained, and the entrywise error decays as $O(n \cdot 2^{-b}) = O(n/2)$ — slowly. The corresponding magic power should match the guesstimate (66).

For the Chebyshev MPO at the same bond dimension $K + 1 = 2$, the error decays super-exponentially, and the MPO captures the effect of *all* R_k gates to within this tolerance. The difference

$$\Delta m = m_{\mathrm{lin}}^{\mathrm{Chebyshev}} - m_{\mathrm{lin}}^{\mathrm{AQFT}} \tag{68}$$

quantifies the contribution of the long-range gates $(R_3, R_4, \ldots)$ to the magic density.

9.3 Hierarchy of approximations

We can organize the approximations as follows:

Method	Bond dim. r	Interp. type	Error scaling
AQFT, $b = 1$	2	Piecewise constant	$O(n/2)$
AQFT, $b = 2$	4	Piecewise constant	$O(n/4)$
Chebyshev, $K = 1$	2	Polynomial	$O(n \cdot 0.65)$
Chebyshev, $K = 3$	4	Polynomial	$O(n \cdot 3.3 \times 10^{-3})$
Chebyshev, $K = 6$	7	Polynomial	$O(n \cdot 2.6 \times 10^{-4})$
Exact QFT	∞	—	0

By computing the transfer-matrix eigenvalues at each level, one obtains a systematic expansion of the magic density in terms of the interpolation quality.

10 Explicit Calculation: $K = 1$ Transfer Matrix

We now carry out the construction explicitly for $K = 1$, which is the simplest nontrivial case.

10.1 Setup

With $K = 1$, we have $r = 2$, and each replica bond index $\alpha_j \in \{0, 1\}$. The composite index $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) \in \{0, 1\}^4$ labels a 16-dimensional space (for the bra) and similarly for $\boldsymbol{\beta}$. Together with the conjugate indices $\boldsymbol{\alpha}', \boldsymbol{\beta}'$, the full transfer matrix is 256×256.

However, by restricting to the S_4-symmetric sector (which is the relevant sector since Q and T_π commute with replica permutations), we can reduce to a much smaller effective dimension.

10.2 Symmetric sector

The symmetric subspace of $\{0, 1\}^4$ under S_4 is spanned by the five "occupation number" states:

$$|m\rangle_{\text{sym}} = \frac{1}{\sqrt{\binom{4}{m}}} \sum_{\substack{\boldsymbol{\alpha} \in \{0,1\}^4 \\ |\boldsymbol{\alpha}| = m}} |\boldsymbol{\alpha}\rangle \,, \qquad m = 0, 1, 2, 3, 4 \,. \tag{69}$$

The transfer matrix restricted to this sector is 25×25 (five choices each for the bra and ket bond).

10.3 Structure of the reduced transfer matrix

In the symmetric sector, the transfer matrix elements take the form

$$\overline{\mathbb{T}}_\pi^{(m,m')(p,p')} = \sum_{\substack{|\boldsymbol{\alpha}|=m,\ |\boldsymbol{\alpha}'|=m' \\ |\boldsymbol{\beta}|=p,\ |\boldsymbol{\beta}'|=p'}} \frac{\mathbb{T}_\pi^{(\boldsymbol{\alpha}\boldsymbol{\alpha}')(\boldsymbol{\beta}\boldsymbol{\beta}')}}{\sqrt{\binom{4}{m}\binom{4}{m'}\binom{4}{p}\binom{4}{p'}}} \,. \tag{70}$$

This 25×25 matrix can be computed analytically by enumerating the physical index sums.

10.4 Computation of individual matrix elements

For each pair of symmetrized bond labels (m, m') and (p, p'), the transfer matrix element involves summing over all physical configurations $(\boldsymbol{\sigma}, \boldsymbol{\tau}, \boldsymbol{\sigma}', \boldsymbol{\tau}')$ weighted by the Q-projector and the permutation.

Concretely, using the Pauli decomposition of Q:

$$\overline{\mathbb{T}}_{\mathrm{id}}^{(m,m')(p,p')} = \frac{1}{16} \sum_{P,P' \in \{I,X,Y,Z\}} \sum_{\sigma,\tau,\sigma',\tau' \in \{0,1\}} \langle \sigma' | P' | \sigma \rangle^4 \, \langle \tau | P | \tau' \rangle^4 \times \overline{A_{p'}^{*m'}(\sigma', \tau')} \, \overline{A_p^m(\sigma, \tau)} \,, \tag{71}$$

where $\overline{A_p^m(\sigma, \tau)}$ denotes the symmetrized version of the product $\prod_{j=1}^4 A^{\alpha_j \beta_j}(\sigma, \tau)$ averaged over all configurations with $|\boldsymbol{\alpha}| = m$ and $|\boldsymbol{\beta}| = p$. This equals

$$\overline{A_p^m(\sigma, \tau)} = \binom{4}{m}^{-1/2} \binom{4}{p}^{-1/2} \sum_{\substack{|\boldsymbol{\alpha}|=m \\ |\boldsymbol{\beta}|=p}} \prod_{j=1}^4 A^{\alpha_j \beta_j}(\sigma, \tau) \,. \tag{72}$$

Using the explicit values from (57), these sums can be evaluated for each (σ, τ), yielding the 25×25 matrix in closed form.

11 Physical Interpretation and Discussion

11.1 Magic from long-range gates

The MPO framework provides a clean separation between the contributions of local (short-range) and nonlocal (long-range) gates to the magic power. At bond dimension $r = 2$ ($K = 1$), the Chebyshev MPO already captures the leading contribution from all gates, while the AQFT at $b = 1$ (same bond dimension) only includes the R_2 gates. The difference quantifies the nonlocal magic contribution.

The numerical data from the working notes shows that the guesstimate $\overline{M}_2 \approx 0.656(N-1)$ underestimates the true magic for $N \gtrsim 5$, suggesting that the long-range gates $(R_3, R_4, \ldots)$ contribute positively to the magic. In the transfer-matrix language, this manifests as $\lambda_{\max}^{\mathrm{Chebyshev}} > \lambda_{\max}^{\mathrm{AQFT}}$ at the same bond dimension.

11.2 Extensivity of magic

The numerical data suggests that $\overline{M}_2(U^{\mathrm{FT}}(N))$ grows linearly with N, consistent with the transfer-matrix prediction of a well-defined magic density. The fitted slopes from the numerical data are:

Input state	Slope (numerics)	Interpretation
Computational basis ($s = 1$)	≈ 0.20	$\log_2(112 - 64\sqrt{3})$ (exact)
Clifford state ($s = 2$)	≈ 0.70	Non-stabilizing power $\overline{M}_2$
Haar random ($s = 3$)	≈ 1.0	Saturates $n - O(1)$

The $s = 1$ result has the exact analytical form derived in the notes, confirming the consistency of the framework.

11.3 Open questions

1. **Exact diagonalization of $\mathbb{T}$ at small K.** Computing the eigenvalues of the 256×256 transfer matrix at $K = 1$ analytically (exploiting the S_4 symmetry reduction) would yield the exact magic density in the Chebyshev-$K = 1$ approximation.

2. **Convergence with K.** How quickly does the magic density converge as K increases? Given the super-exponential convergence of the MPO error, one expects rapid convergence, but the nonlinear dependence of the traces on U could slow this.

3. **Non-linear magic $\overline{M}_2$.** The logarithmic magic $\overline{M}_2 = -\log_2(\ldots)$ is more natural physically but harder to handle analytically, as it involves a Clifford average of a nonlinear function. The transfer-matrix approach directly gives M_{lin}; extending to $\overline{M}_2$ requires either the replica trick or additional machinery.

4. **Dual-unitary circuits.** The formulas developed here for generic unitaries can be specialized to dual-unitary circuits, where additional constraints may further simplify the transfer matrix.

5. **Entanglement cooling transitions.** The MPO formalism naturally connects to the study of entanglement phase transitions under interleaved Clifford and non-Clifford gates, with the QFT providing a concrete non-Clifford layer.

12 Summary and Outlook

We have established a framework for computing the non-stabilizing power of the quantum Fourier transform using the interpolative MPO construction of Chen & Lindsey. The key results are:

(i) The traces $\mathrm{Tr}(Q\,U^{\dagger\otimes 4}\,Q\,U^{\otimes 4}\,T_\pi)$ appearing in the magic power formula can be expressed as transfer-matrix products (33), where the bulk transfer matrix $\mathbb{T}_\pi$ is $r^8 \times r^8$ and *independent of n* due to translational invariance of the MPO cores.

(ii) The thermodynamic limit of the magic density is governed by the leading eigenvalue of $\mathbb{T}_\pi$, per equation (63).

(iii) The AQFT and the Chebyshev MPO fit into the same framework, differing only in the choice of interpolation scheme. This provides a systematic way to quantify the magic contribution of long-range gates.

(iv) For $K = 1$ (bond dimension 2), the problem reduces to diagonalizing a 25×25 matrix (after S_4 symmetry reduction), which is analytically tractable.

The natural next steps are: (a) carry out the explicit diagonalization at $K = 1$ and compare with the numerical data; (b) compute the transfer-matrix eigenvalues numerically for $K = 2, 3$ and verify convergence; (c) extend the framework to the logarithmic magic $\overline{M}_2$ using replica methods.

References

[1] J. Chen and M. Lindsey, "Direct interpolative construction of the discrete Fourier transform as a matrix product operator," *Appl. Comput. Harmon. Anal.* **81**, 101817 (2026).

[2] J. Chen, E. M. Stoudenmire, and S. R. White, "Quantum Fourier transform has small entanglement," *PRX Quantum* **4**, 040318 (2023).

[3] L. Leone, S. F. E. Oliviero, and A. Hamma, "Stabilizer Rényi entropy," *Phys. Rev. Lett.* **128**, 050402 (2022).

[4] D. Coppersmith, "An approximate Fourier transform useful in quantum factoring," arXiv:quant-ph/0201067 (2002).

[5] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge Univ. Press, 2010).

[6] L. N. Trefethen, *Approximation Theory and Approximation Practice*, Extended Edition (SIAM, 2019).

[7] Y. Nam, Y. Su, and D. Maslov, "Approximate quantum Fourier transform with $O(n\log(n))$ T gates," *npj Quantum Inf* **6**, 26 (2020).

[8] P. Hauke *et al.*, "Non-stabilizerness of quantum circuits," arXiv:2505.14793 (2025).

A Explicit Pauli Matrix Elements

For reference, the action of single-qubit Pauli operators in the computational basis:

$$\langle 0|I|0\rangle = 1\,, \quad \langle 0|I|1\rangle = 0\,, \quad \langle 1|I|0\rangle = 0\,, \quad \langle 1|I|1\rangle = 1\,, \tag{73}$$
$$\langle 0|X|0\rangle = 0\,, \quad \langle 0|X|1\rangle = 1\,, \quad \langle 1|X|0\rangle = 1\,, \quad \langle 1|X|1\rangle = 0\,, \tag{74}$$
$$\langle 0|Y|0\rangle = 0\,, \quad \langle 0|Y|1\rangle = -\mathrm{i}\,, \quad \langle 1|Y|0\rangle = \mathrm{i}\,, \quad \langle 1|Y|1\rangle = 0\,, \tag{75}$$
$$\langle 0|Z|0\rangle = 1\,, \quad \langle 0|Z|1\rangle = 0\,, \quad \langle 1|Z|0\rangle = 0\,, \quad \langle 1|Z|1\rangle = -1\,. \tag{76}$$

In the four-replica space, the matrix elements of $P^{\otimes 4}$ are:

$$\langle \boldsymbol{\sigma}|P^{\otimes 4}|\boldsymbol{\sigma}'\rangle = \prod_{j=1}^{4}\langle \sigma_j|P|\sigma_j'\rangle\,. \tag{77}$$

B Q–T_π Trace Identities

For any operator O on $\mathcal{H}^{\otimes 4}$, the traces $\mathrm{Tr}(O\,Q\,T_\pi)$ are constrained by the identity $[Q, T_\pi] = 0$. The 24 permutations in S_4 fall into 6 equivalence classes with respect to Q-twisted traces:

$$\mathrm{Tr}(OQ) = \mathrm{Tr}(OQT_{(12)(34)}) = \mathrm{Tr}(OQT_{(13)(24)}) = \mathrm{Tr}(OQT_{(14)(23)})\,, \tag{78}$$
$$\mathrm{Tr}(OQT_{(24)}) = \mathrm{Tr}(OQT_{(1432)}) = \mathrm{Tr}(OQT_{(13)}) = \mathrm{Tr}(OQT_{(1234)})\,, \tag{79}$$
$$\mathrm{Tr}(OQT_{(14)}) = \mathrm{Tr}(OQT_{(23)}) = \mathrm{Tr}(OQT_{(1342)}) = \mathrm{Tr}(OQT_{(1243)})\,, \tag{80}$$
$$\mathrm{Tr}(OQT_{(12)}) = \mathrm{Tr}(OQT_{(34)}) = \mathrm{Tr}(OQT_{(1324)}) = \mathrm{Tr}(OQT_{(1423)})\,, \tag{81}$$
$$\mathrm{Tr}(OQT_{(132)}) = \mathrm{Tr}(OQT_{(234)}) = \mathrm{Tr}(OQT_{(124)}) = \mathrm{Tr}(OQT_{(143)})\,, \tag{82}$$
$$\mathrm{Tr}(OQT_{(123)}) = \mathrm{Tr}(OQT_{(142)}) = \mathrm{Tr}(OQT_{(243)}) = \mathrm{Tr}(OQT_{(134)})\,. \tag{83}$$

These identities reduce the number of independent traces from 24 to 6.

C Chebyshev–Lobatto Interpolation Details

The Chebyshev–Lobatto nodes on $[0,1]$ are $c^\alpha = \frac{1-\cos(\pi\alpha/K)}{2}$ for $\alpha = 0,\ldots,K$. The Lagrange polynomials can be evaluated via the *barycentric formula*:

$$P^\alpha(x) = \frac{w_\alpha/(x-c^\alpha)}{\sum_{\gamma=0}^{K} w_\gamma/(x-c^\gamma)}\,, \tag{84}$$

where the barycentric weights are

$$w_\alpha = (-1)^\alpha \delta_\alpha \,, \qquad \delta_\alpha = \begin{cases} 1/2 & \alpha = 0 \text{ or } \alpha = K \\ 1 & \text{otherwise} \end{cases} . \tag{85}$$

Key properties:

- Partition of unity: $\sum_{\alpha=0}^{K} P^\alpha(x) = 1$ for all x.
- Interpolation: $P^\alpha(c^\beta) = \delta_{\alpha\beta}$.
- The Lebesgue constant satisfies $\Lambda_K \leq 1 + \frac{2}{\pi}\log(K+1)$.

For $K = 1$: $c^0 = 0$, $c^1 = 1$, $P^0(x) = 1 - x$, $P^1(x) = x$.
For $K = 2$: $c^0 = 0$, $c^1 = 1/2$, $c^2 = 1$, and

$$P^0(x) = 2(x - \tfrac{1}{2})(x - 1)\,, \quad P^1(x) = -4x(x-1)\,, \quad P^2(x) = 2x(x - \tfrac{1}{2})\,. \tag{86}$$